

Latent Variables Moderated Mediation Using Correct and Misspecified Models

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Abstract

TODO: Abstract.

Translational Abstract

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Mediation analysis is a well-known procedure in the behavioral and social sciences (Charlton et al., 2021; Hayes, 2018; Pieters, 2017; Preacher et al., 2007). In its basic and standard form, the effect of an independent variable on an outcome is decomposed into a direct effect and an indirect effect transmitted through a mediator, with all variables observed. This basic single mediator model has since been extended, for example by allowing the indirect effect to depend on a moderator and by adjusting for covariates (Edwards & Lambert, 2007; Hayes, 2015; Montoya, 2024; Preacher et al., 2007).

A more fundamental concern arises from measurement. Many constructs in psychology are not directly observed but are measured indirectly through multiple items, that is, they are latent variables (Bollen & Hoyle, 2012). When such constructs are reduced to summary scores (e.g., a sum or mean of the items) and analyzed as though they were observed, measurement error is ignored. This error can bias the indirect effect and inflate its uncertainty (Cole & Preacher, 2014; Hoyle & Kenny, 1999), and it is especially problematic for the mediator, which acts at once as a predictor and an outcome (Gonzalez & MacKinnon, 2021). To address this, structural equation modelling (SEM) with latent variables has been proposed, especially in the context of mediation (Cheung & Lau, 2008; Falk & Biesanz, 2015; Ledgerwood & Shrout, 2011; Montoya, 2024). By estimating structural relationships between common factors, from which indicator-specific unique variance has been partitioned, provided the measurement model is correctly specified (Bollen, 1989; Rhemtulla et al., 2020). The problem is compounded under moderation, where the product term is even less reliable than either of its constituents (Bohrnstedt & Marwell, 1978; Busemeyer & Jones, 1983); yet we also note that latent variables do not necessarily improve power in general, whether for interactions or not (Cham et al., 2012; Fuller, 1987; Ledgerwood & Shrout, 2011; Wang & Rhemtulla, 2021).

Moderation with latent variables is not straightforward in terms of estimation. Indeed, a latent interaction involves the product of two latent variables, which is unobserved and nonnormal even when its constituents are normal (Klein & Moosbrugger, 2000). A number of methods have

been proposed to estimate latent interactions (e.g., Algina & Moulder, 2001; Arminger & Muthén, 1998; Asparouhov & Muthén, 2021; Boker et al., 2023; Bollen, 1995; Cox & Kelcey, 2021; Holst & Budtz-Jørgensen, 2020; Jöreskog & Yan, 1996; Kenny & Judd, 1984; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Marsh et al., 2004; Mooijaart & Bentler, 2010; Mooijaart & Satorra, 2012; Wall & Amemiya, 2000). Most adopt a joint (“system-wide”) estimation strategy, in which the measurement and structural parameters are estimated simultaneously. This is the case of the most known methods: latent moderated structural equations (LMS) and product-indicator (PI). Opposed to these are structural-after-measurement (SAM) approaches (Cox & Kelcey, 2021; Holst & Budtz-Jørgensen, 2020; Ng & Chan, 2020; Rosseel & Loh, 2024; Wall & Amemiya, 2001, 2003), in which the measurement model is estimated first, from the indicators alone, and the structural model is estimated afterward with the measurement parameters held fixed. Most of the abovementioned approaches have so far been rarely applied in practice, most likely held back by limited software, slow estimation, and error-prone implementations. With recent open-source implementations (Holst & Budtz-Jørgensen, 2020; Rosseel et al., 2025; Slupphaug et al., 2024) and non-commercial software (Keller, 2025), this is likely to change.

In this context, a further consideration concerns interpretation. Mediation is a hypothesis about causal effects, and causal interpretation requires identification assumptions that *may* not be verified from the data alone (Imai et al., 2010; Morelli et al., 2026; Rohrer et al., 2022; VanderWeele, 2015). With latent variable models, further caution is needed (Sterner et al., 2024; VanderWeele & Vansteelandt, 2022). We focus on one that is central to interpretation: the estimated latent variables must validly represent the intended constructs (Gonzalez & MacKinnon, 2021). In fact, when they do not, the model is subject to interpretational confounding (Burt, 1976), a discrepancy between the empirical meaning assigned to a latent variable through its estimated measurement parameters and the meaning intended by the researcher. Under system-wide estimation, the measurement and structural parameters are fit jointly, so the estimated loadings, and with them the empirical meaning of a latent variable, can change with the structural model that is imposed. The latent variable is then no longer anchored to its indicators

alone, and inferences about it need not be consistent across structural models. SAM avoids this by estimating the measurement model on its own. Thus, because the structural model cannot then influence it, the estimated latent variables, and their meaning, do not depend on the structure subsequently fit, and interpretational confounding is avoided. Note, however, that this does not imply the structure is correct (VanderWeele & Vansteelandt, 2022). Local SAM (LSAM) is one such method, which was recently extended to interaction and quadratic effects among latent variables (Rosseel et al., 2025), which makes it applicable to latent variable moderated mediation.

To illustrate it, we take a mediation model with three latent variables, X , M , and Y , each with three indicators, and obtain a population covariance matrix from a correctly specified version. We then alter it so that the covariances between the X and M indicators are unequal across the X indicators (3.0, 1.5, and 1.5, with mean held at the implied value of 2.0), changing only their pattern, not the overall X - M strength. To the same data we fit two structural models, one omitting the mediator and one including it, by system-wide SEM and by SAM ($N = 200$), and compare the estimated loadings of X . As shown in Table 1, the loadings of X estimated by system-wide SEM depend on the structural model: once the mediator enters, the loadings of x_2 and x_3 fall below their true value of one, even though the measurement of X is unchanged, so the empirical meaning of X shifts with the structure imposed. SAM estimates the measurement model from the indicators alone, so the X loadings are identical with and without the mediator and stay at their true values.

The example involves a single, deliberate change to the structural model, but it reflects a general property of system-wide estimation. That is, because the measurement and structural parameters are estimated jointly, the consequences of a misspecification are not confined to the part of the model in which it occurs (Bainter & Bollen, 2014). Prior simulation work has examined latent interaction estimation mainly under nonnormal predictors (Brandt et al., 2014; e.g., Cham et al., 2012; Lonati et al., 2024; Vieira et al., 2026), and in only one case under a misspecified measurement model (Brandt et al., 2020), but not within (moderated) mediation, despite its popularity in applied research. In fact, only a few studies investigated (moderated)

mediation with latent variables, and those always assumed a correctly specified (and saturated) model (Falk & Biesanz, 2015; Feng et al., 2020). Two questions therefore remain open. The first concerns the robustness of the estimators to structural and measurement misspecification, alone and in combination with nonnormal distributions, which is relevant because the estimators rely on different distributional assumptions (e.g., Kelava et al., 2011; Lonati et al., 2024; Vieira et al., 2026). Connected to this, the second concerns how misspecification in the measurement model may affect the estimates, a question that is particularly relevant in moderated mediation, where the mediator functions simultaneously as an outcome and a predictor, and the effect of interest is a nonlinear (product) function of two structural coefficients.

Present Research and Contributions

This study evaluates LSAM, alongside LMS and a PI approach, for latent variable moderated mediation. We assess the three estimators across a wide range of conditions and on a broad set of performance measures, both detailed in the next section. Thus, this study has three contributions to the literature.

These estimators have not been evaluated under misspecification of the measurement or the structural model in a large setting. Moreover, the LSAM approach has not been assessed for derived quantities such as the imm, which is a nonlinear functions of the structural parameters whose behavior need not match that of their constituents. Moreover, for such quantities inference has relied on the bootstrap (Hayes, 2015; Tibbe & Montoya, 2022), especially in the limited prior work concerning (moderated) mediation with latent variables (Falk & Biesanz, 2015; Feng et al., 2020). In this study, we adopt and evaluate the Monte Carlo method, as it may be accurate enough without the computational cost of the bootstrap, which can itself underperform at small sample sizes.

Simulation Design and Performance Evaluation

In this section we describe the design of the simulation study and the measures used to evaluate performance, following recent recommendations for the conduct and reporting of simulation studies (Morris et al., 2019; Pawel et al., 2025; Siepe et al., 2024).

Population Model and Estimand

We study a moderated mediation model among four latent variables: an exogenous predictor X , a moderator W , a mediator M , and an outcome Y . The structural model is as follows

$$M = a_1X + a_2W + a_3(X \times W) + \zeta_M, \quad Y = bM + c'X + \zeta_Y,$$

The primary estimands are the index of moderated mediation (imm) (Hayes, 2015), i.e., a_3b . The remaining structural coefficients (a_1 , a_2 , b , c') and the twelve freely estimated factor loadings serve as secondary estimands. Because the index is a derived quantity rather than a free parameter, its true value is computed as a_3b implied by the correctly specified model.

Population values were fixed at $a_1 = 0.4$, $a_2 = 0.3$, $b = 0.5$, and $c' = 0.15$, with $\text{cor}(X, W) = 0.3$. The interaction coefficient a_3 is a design factor (see below). These coefficients, together with the structural disturbance variances, were chosen so that under the baseline condition (normal exogenous predictors and a correctly specified model) each dependent latent variable has $R^2 = 0.30$ at the interaction level $a_3 = 0.20$. Note that, because a_3 contributes to the variance of M , the explained variance is naturally higher at $a_3 = 0.40$ and lower at $a_3 = 0$. Each latent variable is measured by four congeneric indicators with loadings 1.0, 0.8, 0.7, and 0.6. Figure 1 depicts the population model, where the structural misspecifications studied below are drawn in grey.

Simulation Conditions

We employed a fully factorial design crossing five factors: sample size ($N \in \{100, 200, 300, 500\}$), interaction magnitude ($a_3 \in \{0, 0.20, 0.40\}$), measurement reliability (low = 0.5, high = 0.7), the distribution of the exogenous latent predictors (normal, uniform, t , and two right-skewed χ^2 distributions), and the (mis)specification condition (nine levels; see below). This yielded $4 \times 3 \times 2 \times 5 \times 9 = 1,080$ conditions, each replicated 800 times.

Sample sizes span the small-to-moderate range typical of applied moderated mediation research, where interaction terms are estimated with limited power (Hayes, 2015; Preacher et al., 2007). The interaction magnitudes include a null condition ($a_3 = 0$) to assess Type I error and

false-positive coverage of the index, and two nonzero levels representing small and moderate latent moderation effects. Reliability levels of 0.5 and 0.7 were chosen because measurement error is amplified in product terms and is therefore especially consequential for latent interaction estimation (Bollen, 1989; Brandt et al., 2014), and because lower reliability is common in applied settings than the values typically examined in prior work. Nonnormality of the latent predictors is both empirically common and methodologically central for latent interaction estimators, whose distributional assumptions differ across approaches (Cham et al., 2012; Klein & Moosbrugger, 2000; Klein & Muthén, 2007). The misspecification conditions follow the moderated-mediation framework for the two structural omissions (Preacher et al., 2007) and prior latent-interaction simulation work for the two measurement omissions (Brandt et al., 2020). They were chosen as interpretable threats, acting respectively as a measured mediator–outcome confounder, a conflated moderated direct effect, exposure-correlated measurement error, and an unmeasured residual confounder (Edwards & Lambert, 2007; VanderWeele, 2015). The nine levels are detailed below.

Estimators

We compared a two-step estimator against two system-wide estimators. The two-step approach is LSAM (Rosseel et al., 2025; Rosseel & Loh, 2024). In this case, the interacting predictors X and W shared a single measurement block, while M and Y formed their own blocks. The two system-wide estimators are LMS (Klein & Moosbrugger, 2000) and the extended unconstrained product-indicator approach with double mean centering, which we refer to as UPI throughout (Kelava & Brandt, 2009; Lin et al., 2010; Marsh et al., 2004).

Data Generation

Appendix A describes the complete data-generating mechanism in detail. Briefly, the latent variables X and W were drawn from a bivariate distribution with $\text{Cor}(X, W) = \rho = 0.3$, constructed with the vine-to-anything (VITA) algorithm (Foldnes & Grønneberg, 2015; Grønneberg et al., 2022). Five exogenous distributions were examined, each standardized to mean 0 and variance 1 so that branches differ in shape rather than scale or location: normal, $X, W \sim \mathcal{N}(0, 1)$, using a forced Gaussian copula. The remaining four combine non-normal

margins with a non-Gaussian vine copula: uniform, t_5 , and two $\chi^2(2)$ conditions with X and W skewed in the same or in opposite directions. Given that, M and Y were generated from the structural equations above with independent Gaussian disturbances $\zeta_M \sim \mathcal{N}(0, \psi_M)$ and $\zeta_Y \sim \mathcal{N}(0, \psi_Y)$, and each latent variable's four indicators were formed as $z_{\eta j} = \lambda_j \eta + \delta_{\eta j}$, with loadings $\lambda = (1, 0.8, 0.7, 0.6)$ and Gaussian errors $\delta_{\eta j} \sim \mathcal{N}(0, \theta_{\eta j})$ whose variances $\theta_{\eta j}$ were scaled to the target reliability $\omega \in \{0.5, 0.7\}$.

The nine (mis)specification conditions comprise one correctly specified condition and four families of misspecification, each at two empirically calibrated magnitudes (standardized 0.25 and 0.50, detailed in Appendix A). Two are structural: an omitted direct effect of the moderator on the outcome (c2) and an omitted moderated direct effect (c3). The others are in the measurement model: a cross-loading of a mediator indicator on X (c1) and a residual covariance between a mediator and an outcome indicator (rc). In every case the effect is present in the population but omitted from the fitted model.

Nonconvergence, Outliers, and Inadmissible Solutions

Following previous research (Lonati et al., 2024; Vieira et al., 2026), replications were screened for estimation failures and inadmissible solutions before analysis. Because the compared estimators are implemented across different packages, we applied an identical admissibility check to all of them: a solution was flagged when any residual (unique) variance was nonpositive or when the latent covariance matrix was not positive definite. Each estimator was fit independently within a replication, so a failure of one method produced a missing row for that method only, with the captured error or warning retained for diagnostics.

Performance Measures

Performance was assessed with the measures implemented in the *simhelpers* package (Joshi & Pustejovsky, 2025), following recommendations in Morris et al. (2019). For the index of moderated mediation and the structural coefficients we report relative bias and relative RMSE, the mean standard error, the SE/SD ratio and the relative bias of the variance estimator, confidence-interval coverage, and—where applicable—the rejection rate (Type I error when

$a_3 = 0$, power otherwise). Monte Carlo uncertainty in each performance measure is quantified with Monte Carlo standard errors (Joshi & Pustejovsky, 2025), computed by jackknife where no closed-form expression is available. Table 2 lists the empirical and Monte Carlo standard-error calculations for each measure.

Inference for the index of moderated mediation accounts for the skewness of the product a_3b : for $r = 1, \dots, R$ (with $R = 20,000$) we drew

$$(a_3^{(r)}, b^{(r)}) \sim \mathcal{N}_2((\hat{a}_3, \hat{b}), \hat{\Sigma}_{a_3b}), \quad \text{imm}^{(r)} = a_3^{(r)} b^{(r)},$$

where $\hat{\Sigma}_{a_3b}$ is the estimated covariance of $(\hat{a}_3, \hat{b})$, and took the 2.5th and 97.5th empirical percentiles of $\{\text{imm}^{(r)}\}_{r=1}^R$ as the interval. The structural coefficients and loadings use Wald (?) intervals from the model-implied parameter covariance matrix. The same implementation was applied to all three estimators for comparability.

Reproducibility and Computational Details

All analyses were conducted in R (R Core Team, 2025) using the *lavaan* (Rosseel, 2012) and *modsem* (Slupphaug et al., 2024) packages for estimation, *covsim* (Grønneberg et al., 2022) and *rvinecopulib* (Nagler & Vatter, 2025) for data generation, and *simhelpers* (Joshi & Pustejovsky, 2025) for performance evaluation. Conditions were run in parallel with the *parallel* package. Reproducible and independent random-number streams across workers were obtained with the L'Ecuyer-CMRG generator. The computational environment was pinned with Nix via the *rix* package (Rodrigues & Baumann, 2026) to ensure reproducibility of the computational environment, and the article was written with the *apaquarto* extension (Schneider, 2026). For the package versions, code and materials, see the project repository.

Results

We present the results in three sections. We first report the imm and the interaction coefficient. We then turn to the remaining structural coefficients (i.e., a_1 , a_2 , b , and c'), and we close with the estimated factor loadings, focusing on the cross-loaded indicator. Due to the large number of conditions, the figures show a focused subset of the results. We fix the interaction

coefficient at 0.2 (except for Type I error, which by construction requires a zero interaction) and display a limited number of sample sizes: $N = 200$ and 500 for the imm and the interaction coefficient, and $N = 500$ for the structural coefficients and loadings. For the latter two we further restrict to high reliability, whereas the imm and the interaction coefficient are shown at both reliability levels. The entire set of results, across all sample sizes, reliabilities, and interaction values, is available in the repository.

Main Structural Estimates

Relative Bias

Under correct specification with normal predictors, all three estimators recover both the imm and a_3 with negligible bias (Figure 2). In fact, bias is negligible across the nonnormal predictors as well, with the exception of the χ^2 conditions, where LMS and LSAM are the most affected. In this case, however, larger sample size removes the bias for LSAM but not for LMS, while higher reliability lessens it for both. Under misspecification, both c2 and c3 bias the imm for all three estimators, as expected. For a_3 , c2 produces no bias, whereas c3 leads to bias under the system-wide estimators. Among the measurement omissions, c1 attenuates a_3 only under the system-wide estimators, while for the imm LMS shows bias in the case of χ^2 conditions. Last, the rc leaves a_3 and the imm essentially unchanged. In all cases the bias increases with the magnitude of the omission and at lower reliability.

Note the bias of a_3 under c3 is the one result not implied by the omission alone. As both omissions sit in the outcome equation, the mediator equation that defines a_3 stays correct. Hence, LSAM, which estimates a_3 from the mediator block alone, recovers it. The system-wide estimators fit the full covariance structure, where the latent interaction reaches the outcome only through a_3 and b , and absorb the omitted covariance into a_3 .

Relative RMSE

Under correct specification, relative RMSE falls with sample size and reliability for all three throughout distributions (Figure 3). Specifically, LMS is the most efficient, LSAM intermediate, and UPI the least. However, under misspecification, the structural omissions (c2 and

c3) inflate the RMSE of the imm for all three estimators. In fact, such increase, in some cases, affects the system-wide estimators the most. With respect to the measurement omissions (c1 and rc), we observed a negligible effect. For a_3 , only c3 inflates slightly the RMSE mainly for UPI, while LSAM and LMS are unaffected. The inflation is larger at lower reliability.

Coverage

Under correct specification with normal predictors, coverage is around the nominal level with all three approaches (Figure 4 and Figure 5) for the imm and a_3 . It goes down slightly under heavy-tailed (t_5) and χ^2 predictors for LMS and UPI, respectively, but mainly with a smaller sample size and low reliability.

Under misspecification, c2 and c3 reduce the imm coverage, since its interval is then centered on a biased estimate, with the largest drop for the system-wide estimators under c3. For a_3 , the cross-loading (c1) is the damaging case, collapsing coverage under the system-wide estimators at low reliability while LSAM holds near nominal. The remaining omissions leave coverage close to nominal.

Standard errors

Under correct specification the SE/SD ratio is near one for all three approaches (Figure 6 and Figure 7) in the case of a high reliability. Indeed, an overestimation is observed in case of low reliability and smaller sample size for UPI. Moreover, in the χ^2 in opposite directions, LSAM and UPI struggled the most throughout all misspecifications, including with the correct specification. These results were highly similar for both imm and a_3 . We also observe that the measurement misspecifications affect in particular LMS in combination with specific nonnormal distributions. Similar conclusions are obtained with the relative bias of the variance estimator (Figure 8 and Figure 9).

Under misspecification the SE/SD ratio remains near one for all three. The variance-bias measure stays near one at high reliability but becomes unstable at low reliability, where a few extreme replications inflate it for every method and most severely for LMS, with the largest cells truncated in the figures.

Hypothesis Testing (Type I error and Power)

Under correct specification with normal predictors and $a_3 = 0$, the imm test holds its nominal level reasonably well (Figure 10). UPI is slightly conservative, LSAM is closest to nominal, and LMS is liberal, with the same ordering for a_3 . The inflation of LMS grows under nonnormal predictors, exceeding the nominal level in several skewed and heavy-tailed cells, while LSAM and UPI remain near it.

Under misspecification, c_3 inflates the false-positive rate of the system-wide estimators, LMS the most, while LSAM holds its level. The cross-loading (c_1) likewise inflates LMS, mainly at low reliability. The remaining omissions leave the rate near nominal.

At $a_3 = 0.2$ the index is detected with power that rises with sample size and reliability (Figure 11). LMS has the highest power throughout and UPI the lowest, with LSAM close to LMS, mirroring their efficiency ordering. Power falls under opposite-skew χ^2 predictors, dropping well below the conventional 0.80 at the smaller sample size, and most for UPI and LSAM. We note however that interpreting power in the case of bias and inflated Type I error should be avoided (Morris et al., 2019).

Additional Structural Estimates

For the remaining structural coefficients (a_1 , a_2 , b , and c'), at high reliability and $N = 500$, we report their relative bias (Figure 12), coverage (Figure 13), and SE/SD ratio (Figure 14) across all misspecification conditions and predictor distributions.

The coefficients show where each omission is absorbed. The two structural omissions act on the outcome equation: c_2 biases the mediator-outcome slope b and, more strongly, the direct effect c' , and c_3 is absorbed mainly by b . Under LSAM the mediator-equation coefficients a_1 and a_2 stay essentially unbiased under both structural omissions, since LSAM estimates the mediator block in isolation; the system-wide estimators, by contrast, leak c_2 into the mediator equation, biasing a_2 (about 11% at the medium level) and, to a lesser extent, a_1 , and likewise carry part of c_3 into c' . The cross-loading (c_1) is the most disruptive, biasing all four coefficients for every estimator, because the omitted X loading on m_3 distorts the mediator's relations to the other latent

variables. The residual covariance (rc) slightly inflates b and biases c' downward for all three. Across conditions the bias is consistently smaller under LSAM than under the system-wide estimators.

Taken together, under correct specification the three estimators recover the imm and a_3 with little bias and near-nominal coverage, and differ mainly in a calibration and efficiency trade-off. LMS is the most efficient but runs liberal and is the most sensitive to nonnormality. LSAM is the best calibrated in both coverage and Type I error at a modest efficiency cost. UPI is intermediate but unstable at the smallest sample size.

Measurement Loadings

Finally, for the twelve freely estimated factor loadings, all but one are recovered with near-nominal performance across every measure and condition. Indeed, as expected, the cross-loaded mediator indicator m_3 : under the omitted cross-loading ($c1$) its relative bias rises and its coverage collapses, while the other eleven loadings stay flat throughout. One subtler pattern is worth noting, even though it is negligible in size: under the residual covariance (rc), the system-wide estimators carry a systematic upward bias in m_3 , well under one percent, whereas LSAM stays centered on zero, since it estimates the mediator's measurement block in isolation. We report only m_3 , with its relative bias, coverage, SE/SD ratio, and variance bias shown together (Figure 15) at high reliability and $N = 500$.

Discussion

We evaluated three estimators of a latent variable moderated mediation model, the two-step LSAM and the system-wide LMS and UPI, under correct specification and under a large number of structural and measurement misspecifications. Two questions organized the study: how the estimators recover the index of moderated mediation and its constituents when the model is correct, and how each responds when the fitted model omits an effect that is present in the population.

Under correct specification the three estimators agree on the substance. All recover the imm and the interaction coefficient with negligible bias and near-nominal coverage, and they

separate mainly along a calibration and efficiency trade-off. LMS is the most efficient, with the lowest RMSE throughout, but it runs liberal and is the most sensitive to nonnormal predictors, where its Type I error climbs above the nominal level. Note the bias is dependent on the type of nonnormality, similar to previous findings (Lonati et al., 2024; Vieira et al., 2026). LSAM is the best calibrated in both coverage and Type I error, at a modest cost in efficiency. UPI sits between the two but does not yield usable estimates at the smallest sample size. For a correctly specified model the choice is therefore between efficiency and calibration rather than between right and wrong answers, and it depends on the sample size and on the plausibility of normal predictors.

The estimators diverge under misspecification, and the divergence follows from how each ties the equations together. Under system-wide estimation a misspecification cannot be contained: because all parameters are fit jointly to the full indicator covariance matrix, an omitted effect is reproduced through whatever paths the model makes available and spreads into parameters that bear no structural relation to it, including the interaction coefficient and the loadings, biasing them and inflating the Type I error of the index. Structural omissions mainly perturb the structural estimates, whereas measurement omissions matter most in specific, demanding settings, in particular at low reliability—patterns broadly consistent with previous work (cite?). LSAM avoids this by estimating the measurement model first and reading the structural parameters from blocks that do not involve the misspecified equation; the separation does not make the structural model correct, but it keeps a local error local. This is the interpretational confounding that motivated the study, now visible in a moderated mediation setting and in a derived quantity, the imm.

For practice, the results suggest a pragmatic division. When the measurement and structural models are trusted and the predictors are approximately normal, LMS is an efficient default, with the caveat that its standard errors run optimistic and its Type I error is sensitive to nonnormality. When either assumption is in doubt, and especially when the interaction coefficient or the index of moderated mediation is the target, LSAM is the safer choice, because its estimate of the interaction is insulated from misspecification elsewhere in the structural model and from contamination in the measurement model. UPI is a reasonable middle option at moderate and

large sample sizes but should not be relied on when samples are small. Across estimators, inference for the index by the Monte Carlo method was well calibrated, which supports it as an alternative to the bootstrap for this nonlinear quantity.

The study has limits that bound these conclusions [...]

Several extensions follow naturally [...]

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Table 1*Illustration for Interpretational Confounding.*

	Parameter	True	SEM, no M	SEM, with M	SAM, no M	SAM, with M
x1	Loading of x_1	1	1	1.00	1	1
x2	Loading of x_2	1	1	0.85	1	1
x3	Loading of x_3	1	1	0.85	1	1

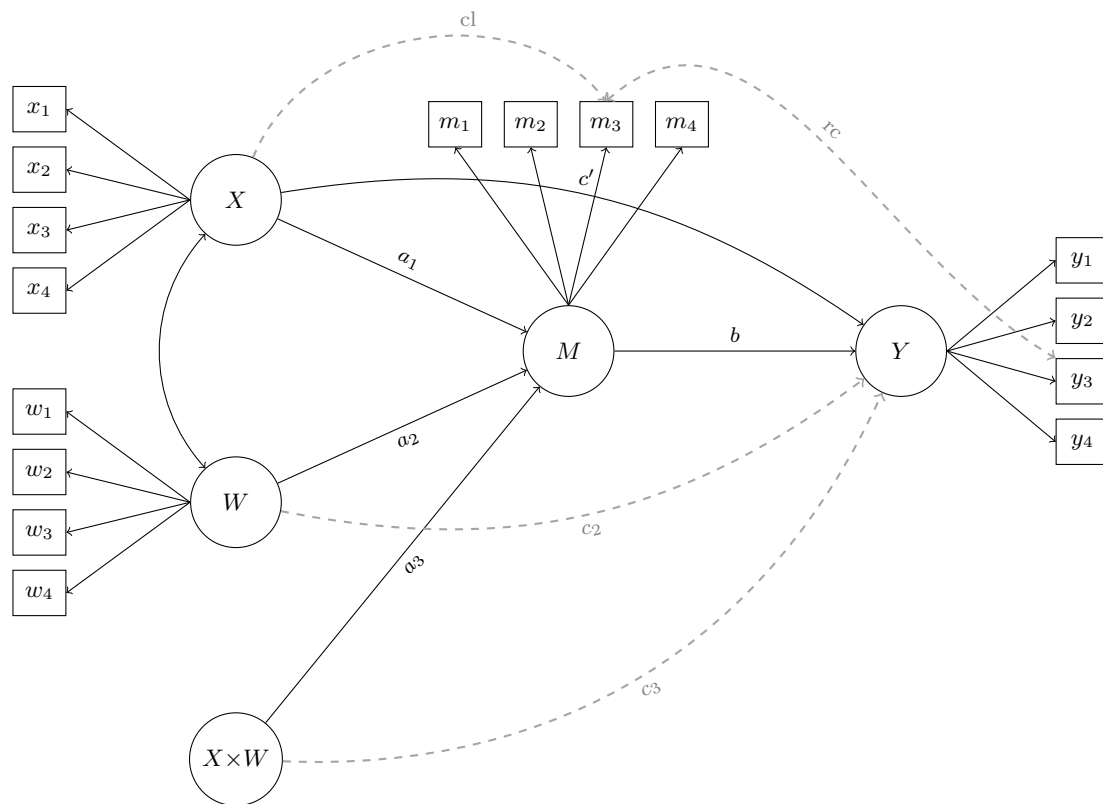
Table 2*Subset of the Performance Measures for Evaluating Parameter Estimates*

Performance measure	Empirical calculation	MCSE
<i>Relative criteria</i>		
Relative bias	$\frac{\bar{B}}{\theta} - 1$	$\sqrt{S_B^2 / (K\theta^2)}$
Relative RMSE	$\frac{1}{ \theta } \sqrt{\frac{1}{K} \sum_{k=1}^K (B_k - \theta)^2}$	$\frac{1}{ \theta } \sqrt{\frac{K-1}{K} \sum_{j=1}^K (\text{RMSE}_{(j)} - \text{RMSE})^2}$
<i>Standard errors</i>		
Mean SE	$\overline{SE} = \frac{1}{K} \sum_{k=1}^K \widehat{SE}_k$	—
SE/SD ratio	$\overline{SE} / S_B$	—
Relative bias of variance estimator	$\rho = \overline{SE^2} / S_B^2$	$\sqrt{\frac{K-1}{K} \sum_{j=1}^K (\rho_{(j)} - \rho)^2}$
<i>Coverage</i>		
Coverage	$c = \frac{1}{K} \sum_{k=1}^K I(L_k \leq \theta \leq U_k)$	$\sqrt{c(1-c)/K}$
<i>Hypothesis testing</i>		
Rejection rate	$r = \frac{1}{K} \sum_{k=1}^K I(0 \notin [L_k, U_k])$	$\sqrt{r(1-r)/K}$

Note. B_k is the estimate of parameter θ in replication k ($k = 1, \dots, K$ usable replications), $\bar{B}$ its mean, S_B^2 its sample variance, and S_B its standard deviation; $\text{RMSE} = \sqrt{\frac{1}{K} \sum_{k=1}^K (B_k - \theta)^2}$; $\widehat{SE}_k$ is the estimated standard error, $\overline{SE^2} = \frac{1}{K} \sum_{k=1}^K \widehat{SE}_k^2$, and L_k, U_k the confidence-interval limits; c and r denote coverage and rejection rate, and ρ the relative bias of the variance estimator. The subscript (j) marks a leave-one-out jackknife replicate. The MCSEs for relative RMSE and the relative bias of the variance estimator are computed by jackknife; the others are closed-form, and dashes mark MCSEs that are not computed. All measures were computed with the *simhelpers* package.

Figure 1

Population (Data-Generating) Model for the Simulation Study.



Note. X , W , M , and Y are latent variables, each measured by four congeneric indicators; circles are latent variables and $X \times W$ is the latent interaction. Black elements form the correctly specified model that every estimator fits (a_1 , a_2 , a_3 , b , c' , and the X – W covariance). Grey elements are the misspecifications present in the population but omitted from the fitted model: two in the structural part, an omitted moderator effect on the outcome (c_2) and an omitted moderated direct effect (c_3); and two in the measurement part, an omitted cross-loading of X on m_3 (c_1) and an omitted residual covariance between m_3 and y_3 (rc). Residual variances and intercepts are not shown.

Figure 2
Relative Bias of the Index of Moderated Mediation (imm) and the Interaction Coefficient a_3

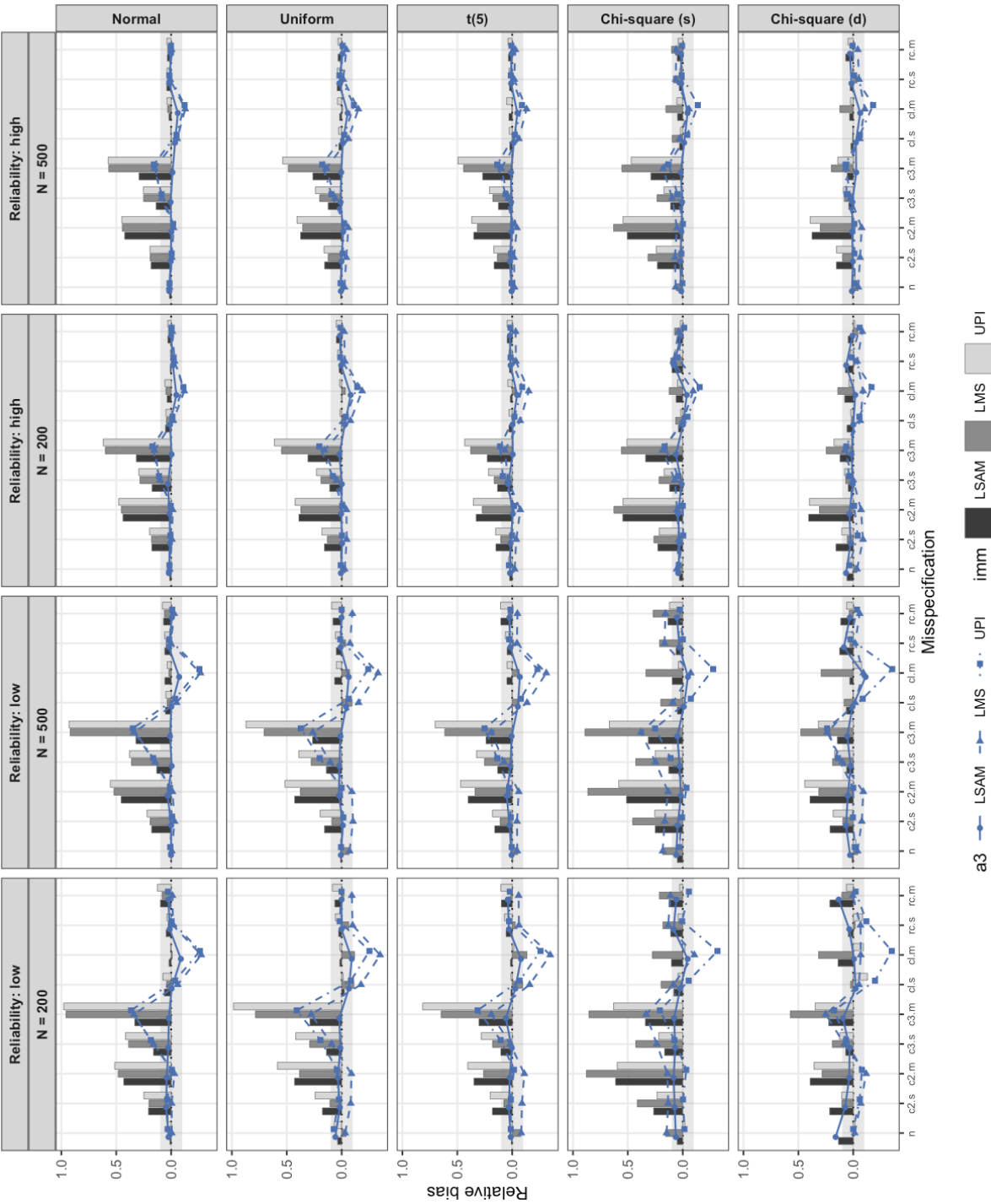


Figure 3

Relative RMSE of the Index of Moderated Mediation (imm) and the Interaction Coefficient a_3

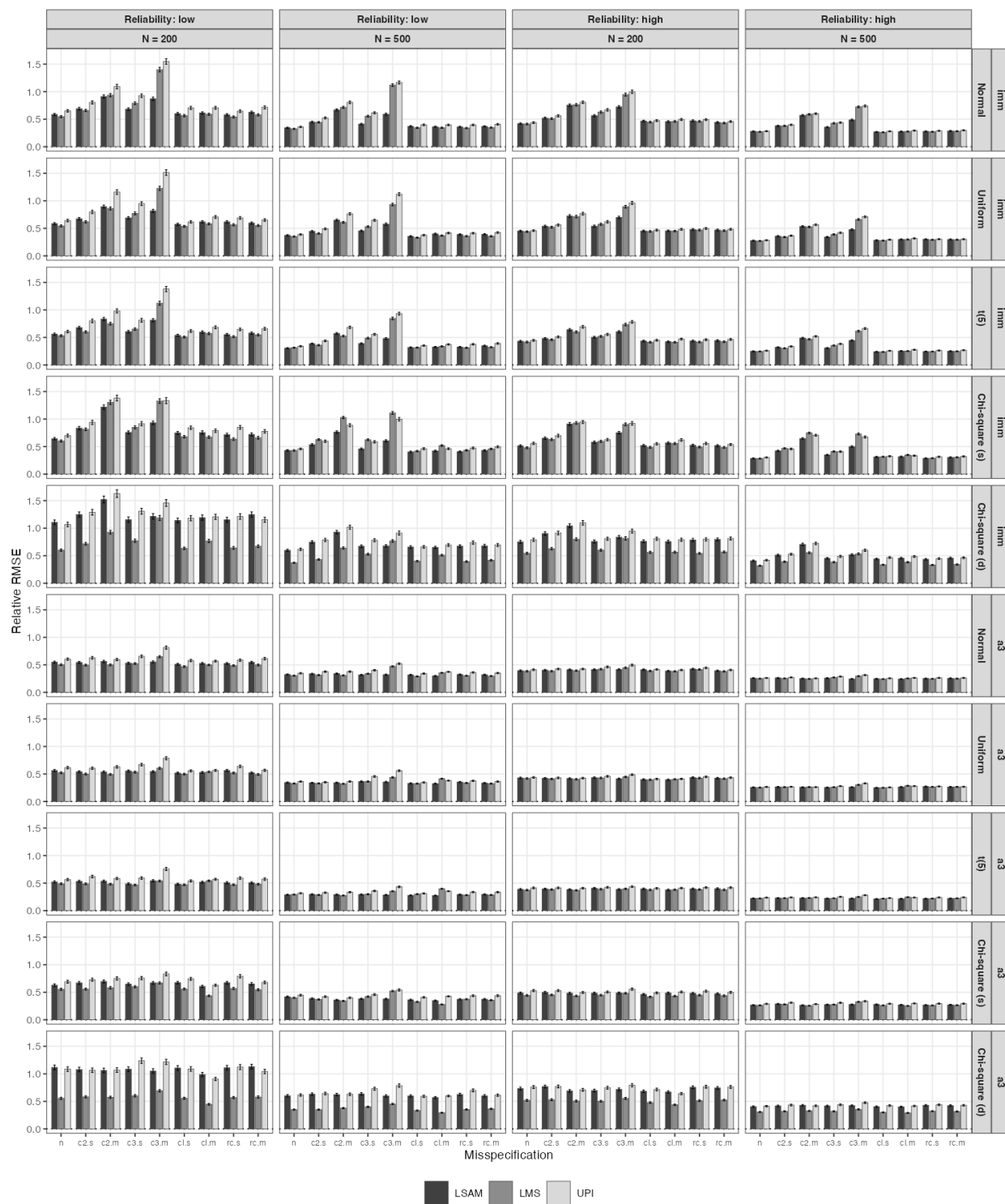


Figure 4
Coverage of the Index of Moderated Mediation

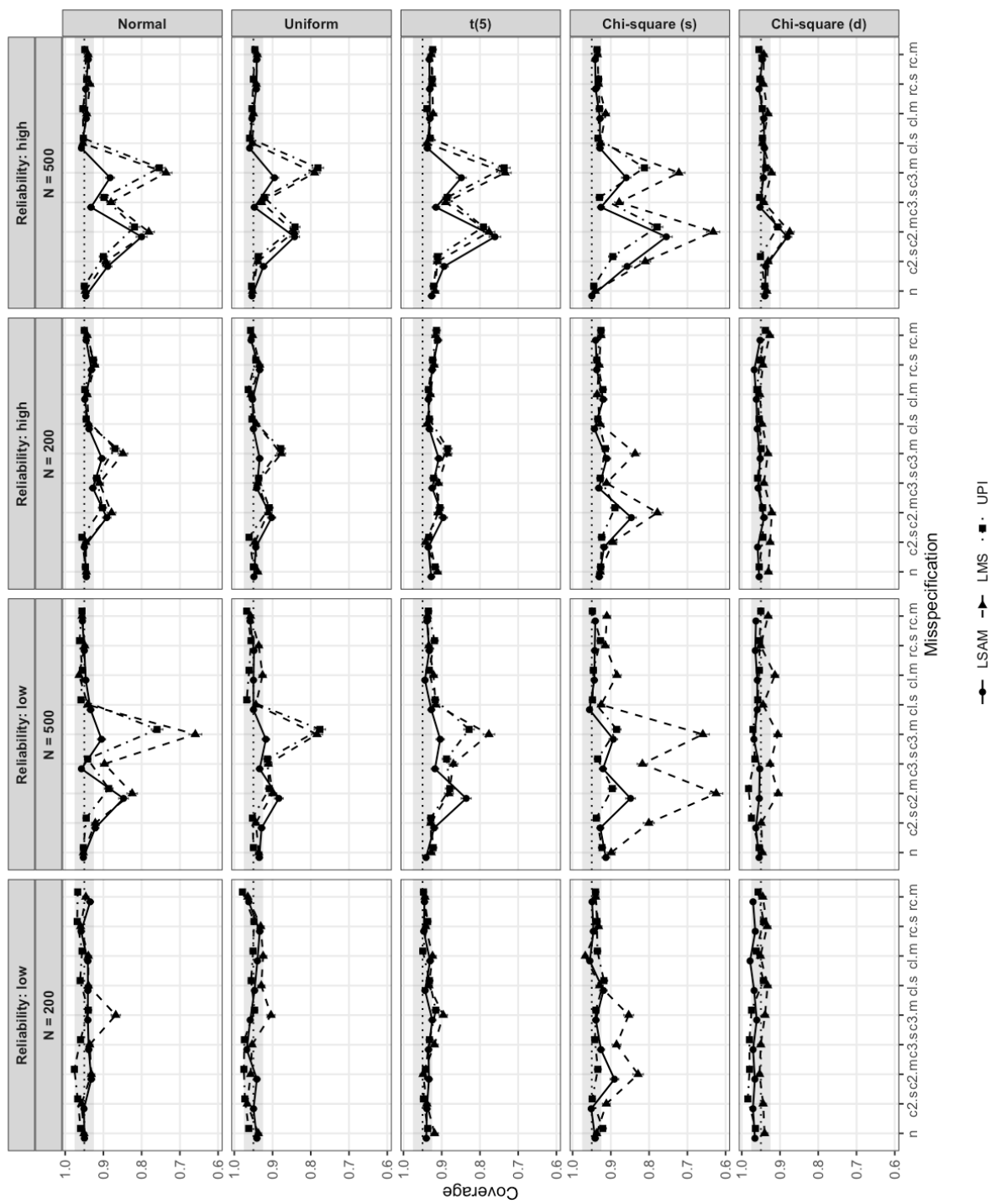


Figure 5
Coverage of the Interaction Coefficient a_3

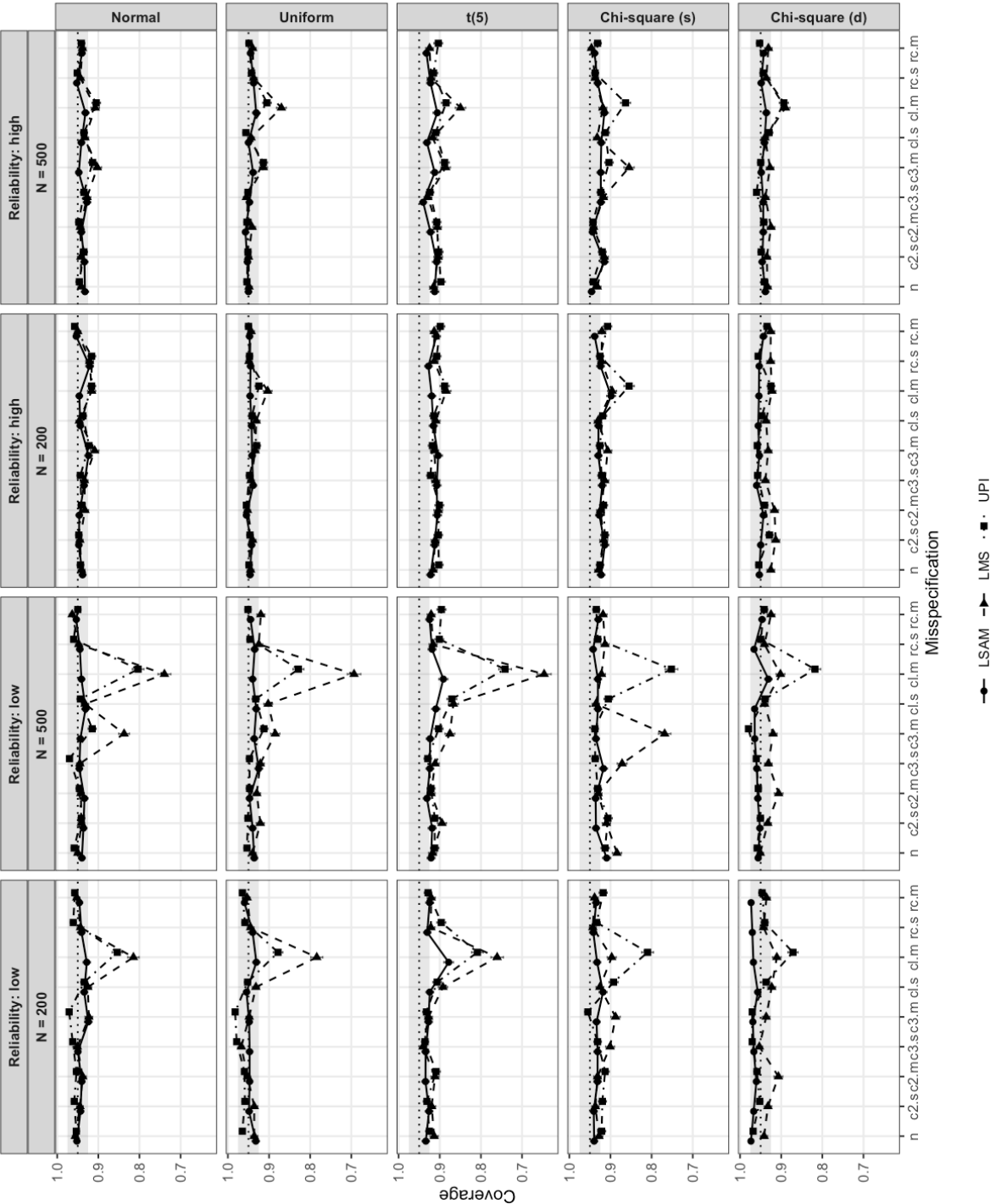


Figure 6
SE/SD of the Index of Moderated Mediation

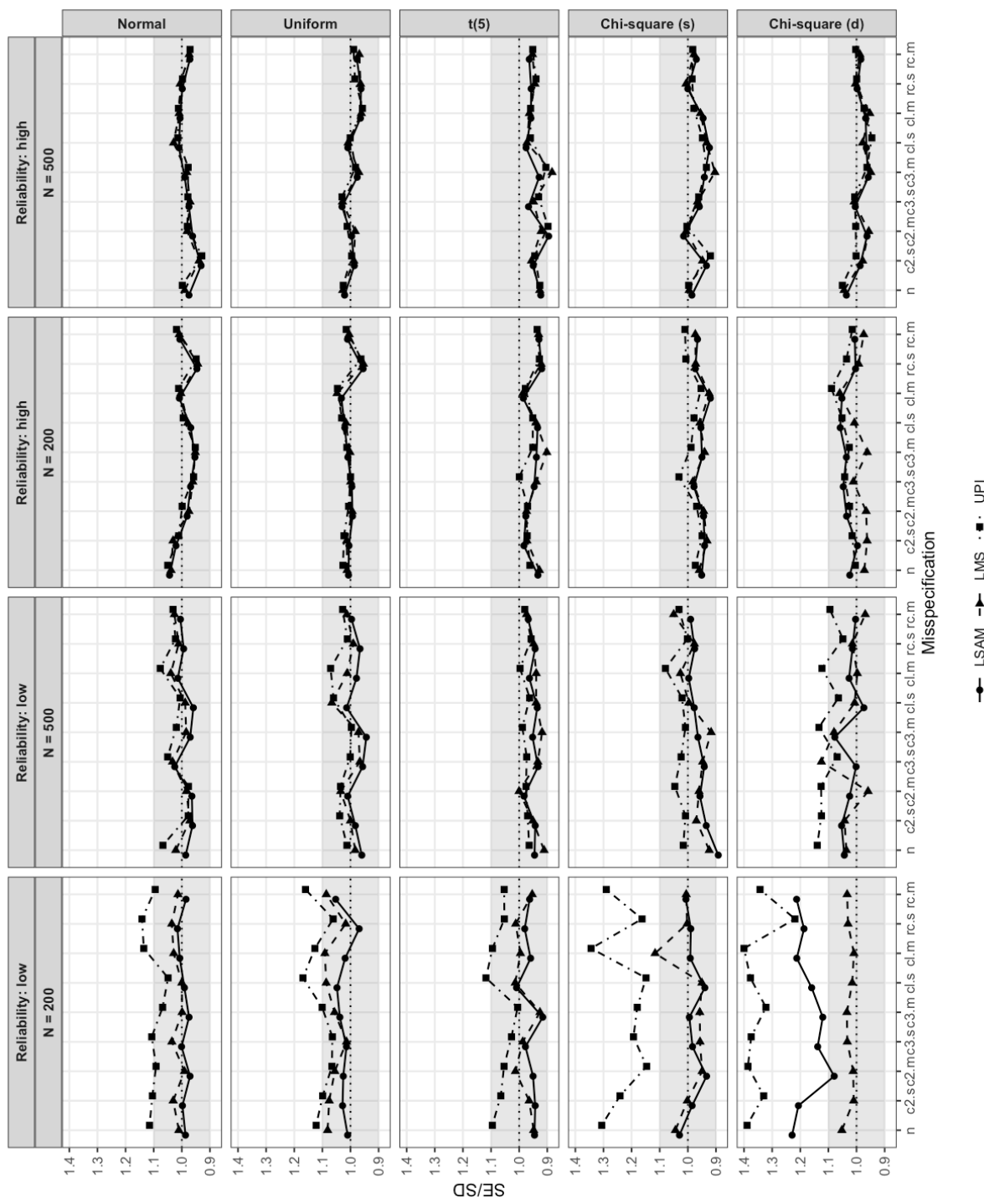


Figure 7
SE/SD of the Interaction Coefficient a_3

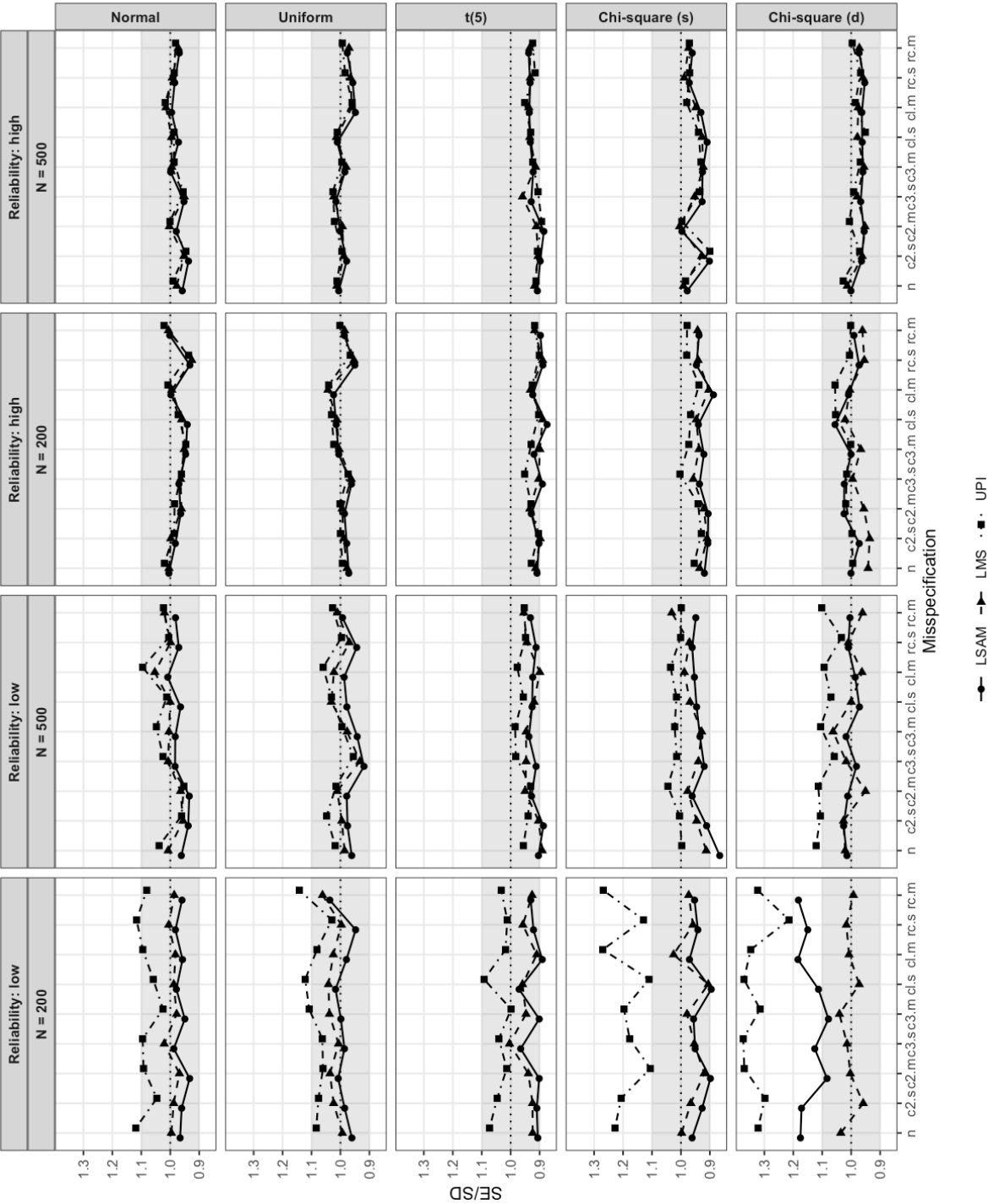


Figure 8
Relative Bias of the Variance Estimator of the Index of Moderated Mediation

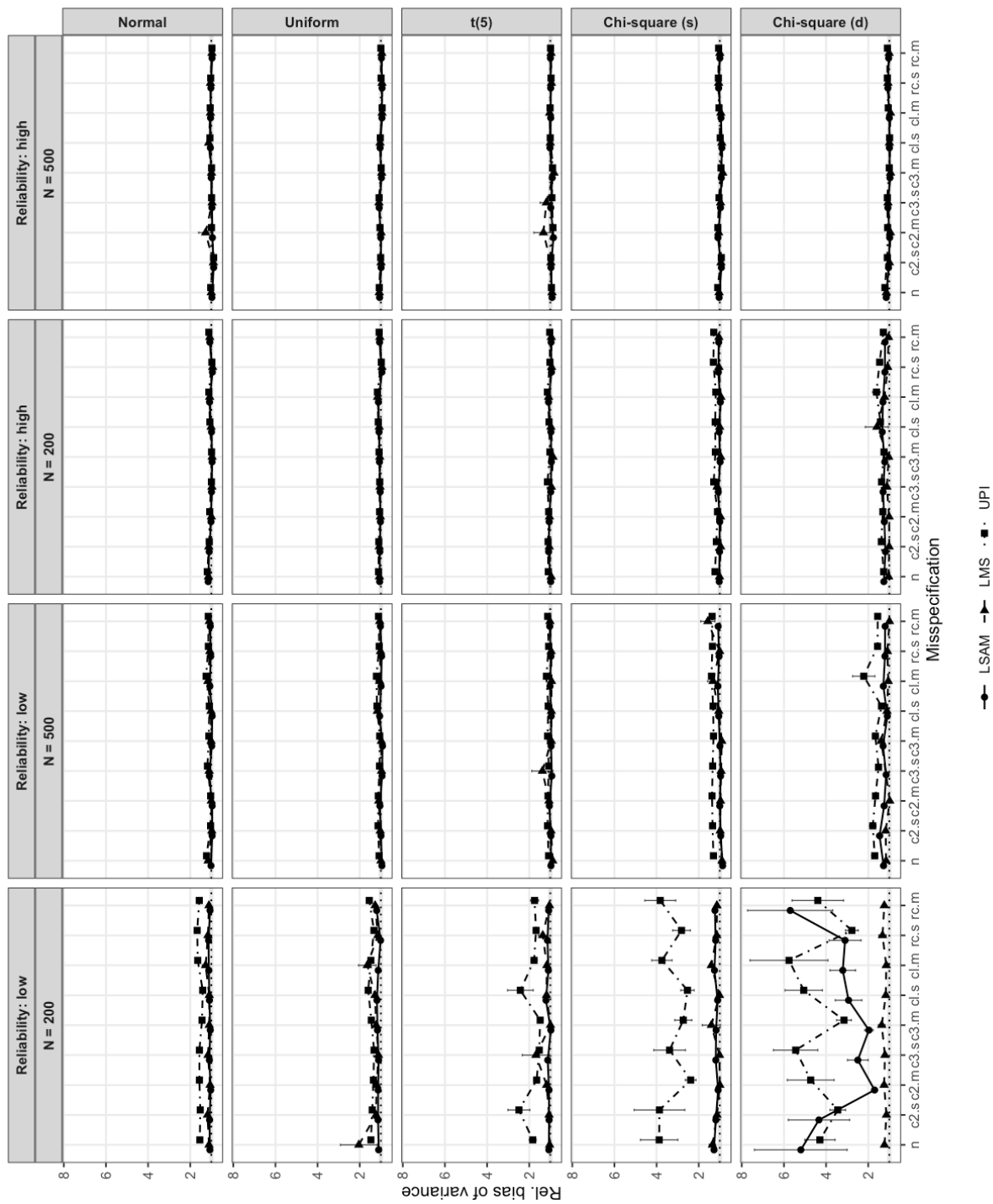


Figure 9
Relative Bias of the Variance Estimator of the Interaction Coefficient a_3

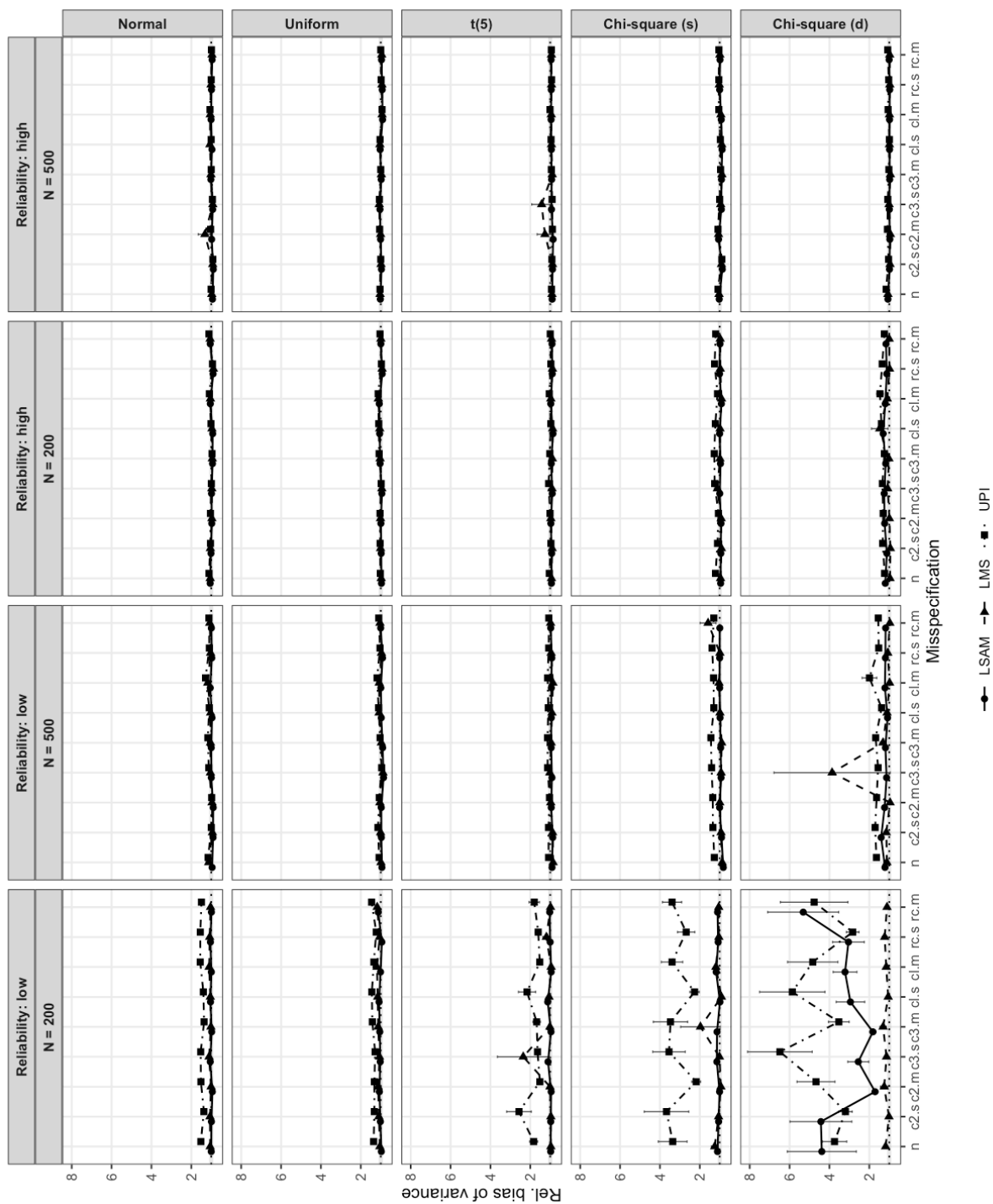


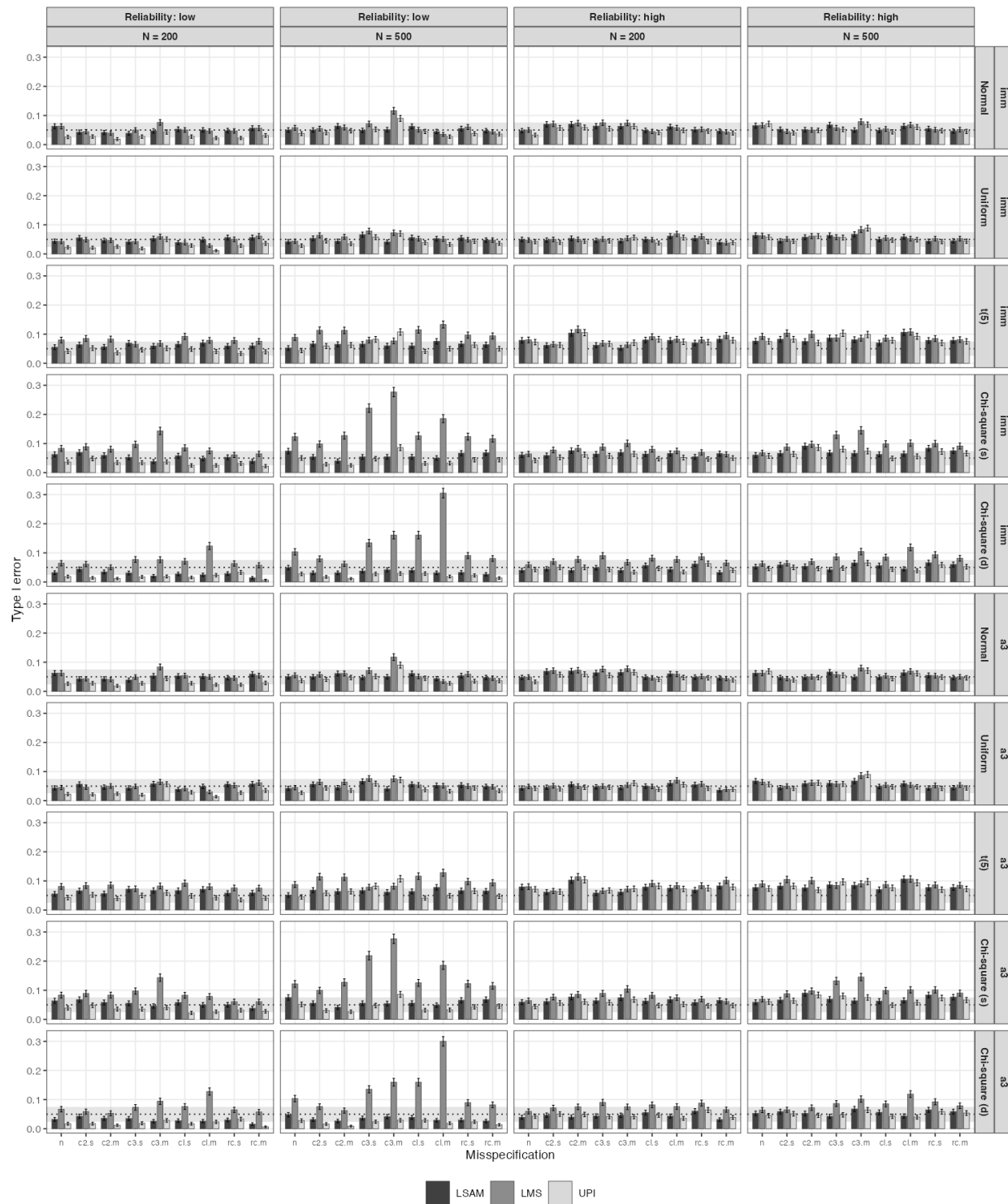
Figure 10*Type I Error of the Index of Moderated Mediation (imm) and the Interaction Coefficient a_3* 

Figure 11

Power for the Index of Moderated Mediation (imm) and the Interaction Coefficient a_3 at $a_3 = 0.2$

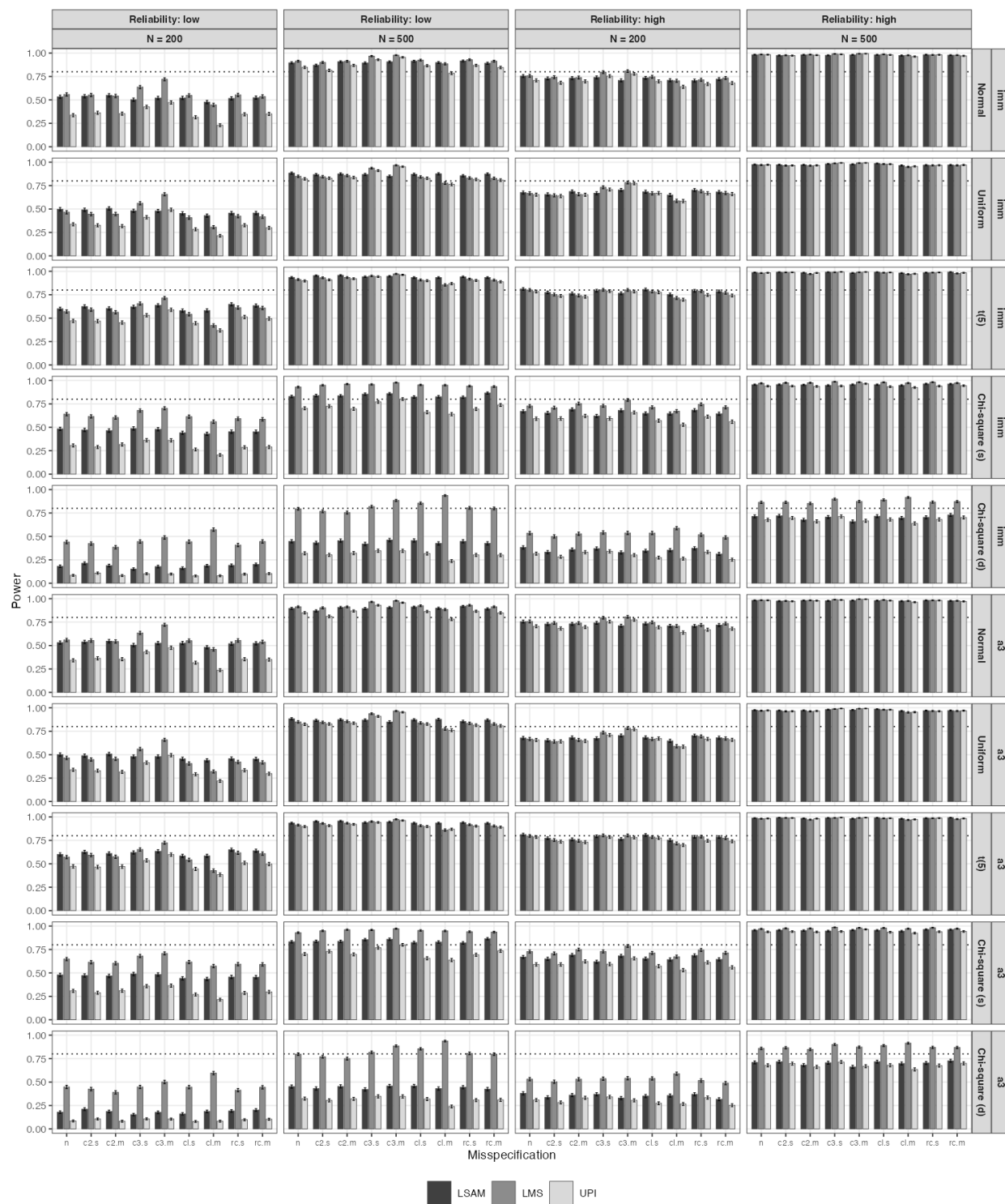


Figure 12
Relative Bias of the Structural Coefficients (a_1 , a_2 , b , c') at High Reliability and $N = 500$

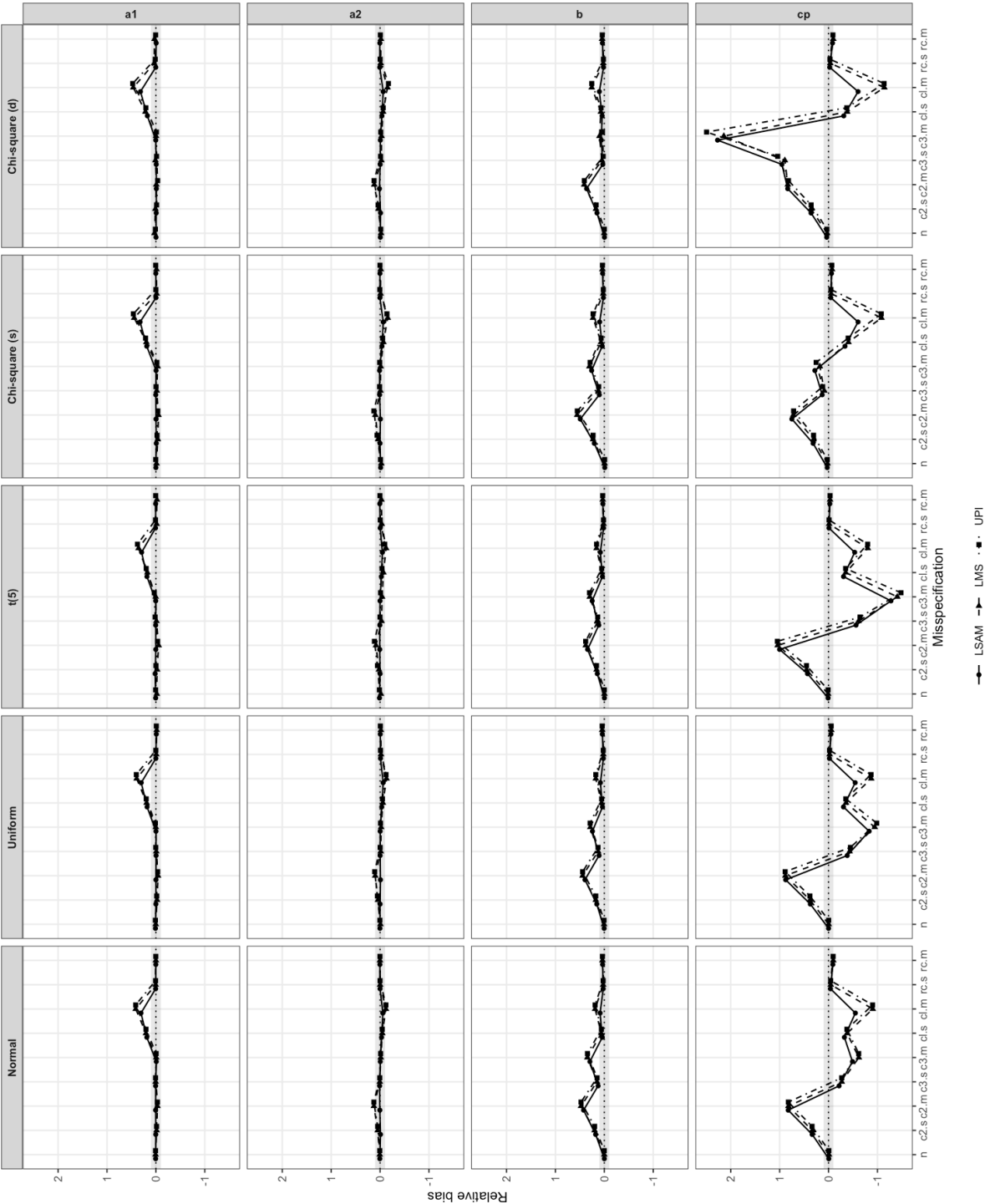


Figure 13
Coverage of the Structural Coefficients (a_1 , a_2 , b , c') at High Reliability and $N = 500$

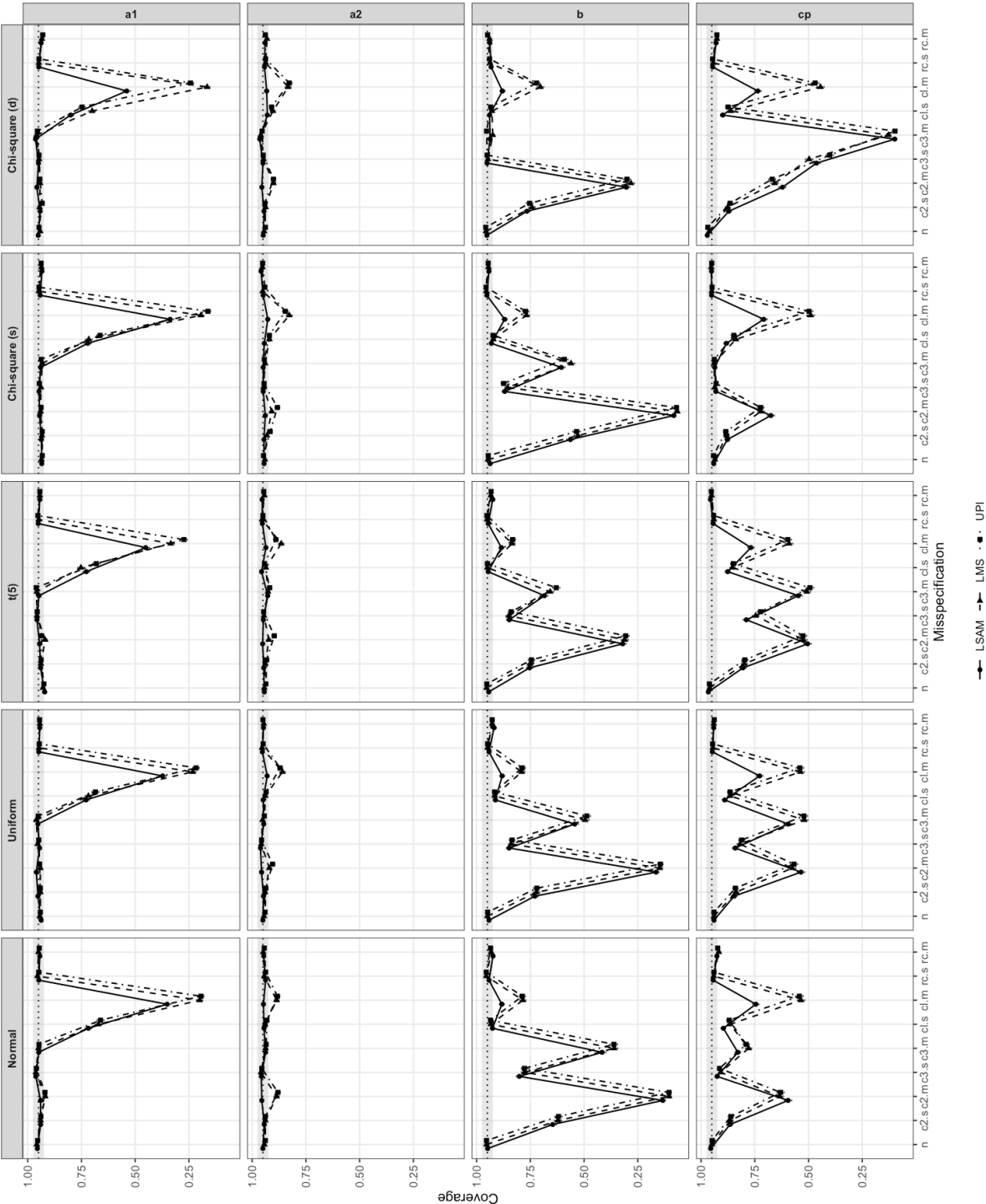


Figure 14
SE/SD of the Structural Coefficients (a_1 , a_2 , b , c') at High Reliability and $N = 500$

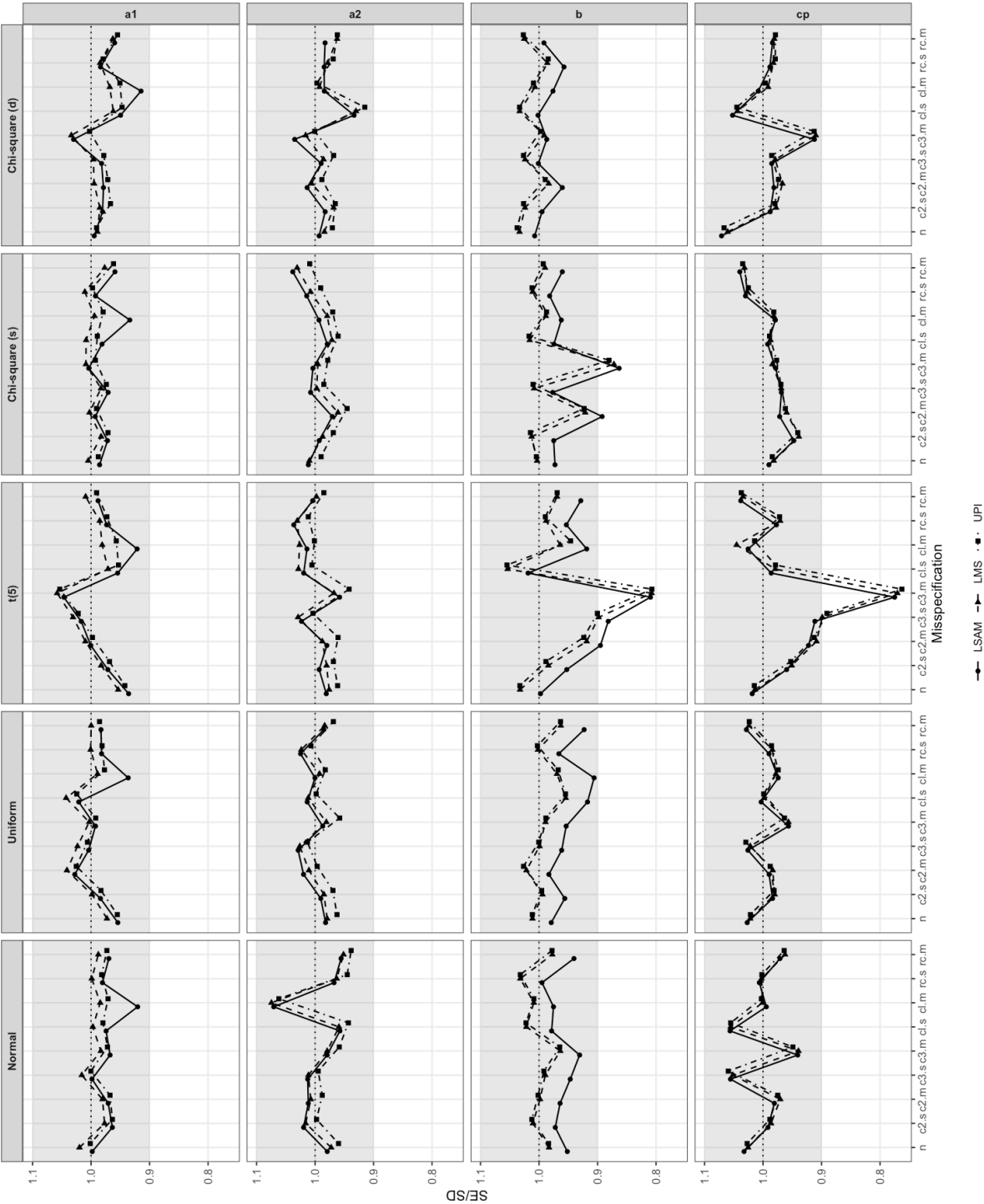
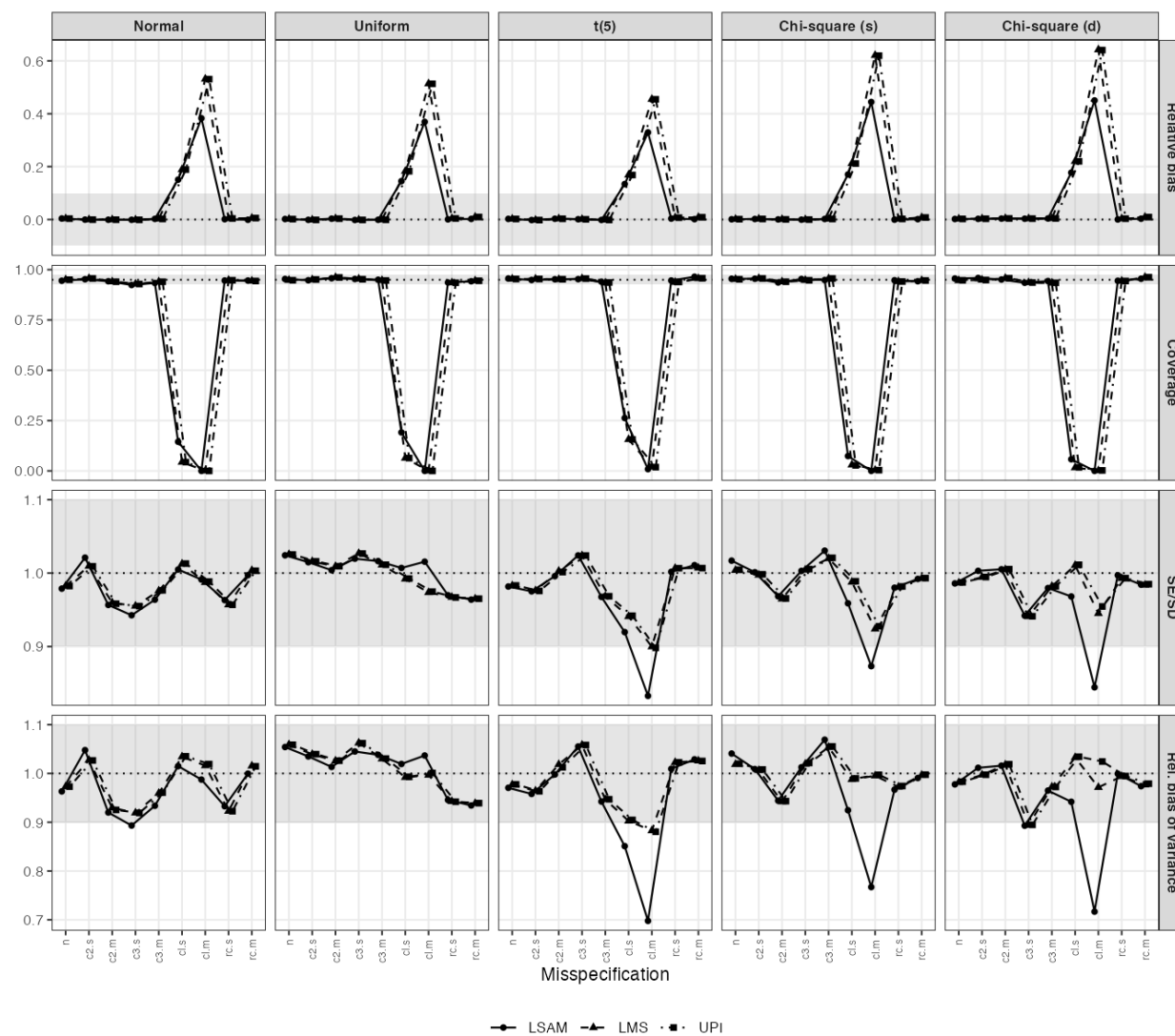


Figure 15

Performance of the Cross-Loaded Mediator Indicator Loading (m_3) Across All Measures at High Reliability and $N = 500$



Appendix

Data Generation and Calibration

This appendix gives the formulas behind the data-generating mechanism. The full derivations, with worked numerical examples, are in the project repository.

Structural residual variances

The disturbance variances ψ_M, ψ_Y are fixed so that each structural equation explains a fraction $R^2 = 0.30$ of its outcome's variance at the central interaction level $a_3 = 0.2$ under the normal baseline. Writing the systematic part of M as $S_M = a_1X + a_2W + a_3(XW)$, its variance needs only second moments,

$$\text{Var}(S_M) = a_1^2 + a_2^2 + 2a_1a_2\rho + a_3^2(1 + \rho^2),$$

and the disturbance is the remainder that brings the signal share to R^2 ,

$$\psi_M = \text{Var}(S_M) \frac{1 - R^2}{R^2}, \quad \psi_Y = \text{Var}(bM + c'X) \frac{1 - R^2}{R^2},$$

with $\text{Var}(bM + c'X) = b^2 \text{Var}(M) + c'^2 + 2bc'(a_1 + a_2\rho)$. Fixed at these values, the population variances of M and Y are closed-form functions of the parameters,

$$\text{Var}(M) = \text{Var}(S_M) + \psi_M, \quad \text{Var}(Y) = b^2 \text{Var}(M) + c'^2 + 2bc'(a_1 + a_2\rho) + \psi_Y,$$

evaluated once under the normal baseline and held fixed across distributions, so that R^2 and reliability drift only through the design factors.

Indicator reliabilities

Each latent $\eta \in \{X, W, M, Y\}$ has four indicators $z_{\eta j} = \lambda_j \eta + \delta_{\eta j}$ with $\lambda = (1, 0.8, 0.7, 0.6)$. To hit a target reliability $\omega \in \{0.5, 0.7\}$, the residual variance is set from the latent variance $V = \text{Var}(\eta)$,

$$\theta_{\eta j} = \lambda_j^2 V \frac{1 - \omega}{\omega}.$$

For X, W the variance is exactly 1; for M, Y it is the closed form above, fixed at the normal baseline, so the realized reliability drifts mildly across distributions.

Exogenous distributions

In every cell (X, W) has mean 0, variance 1, and correlation $\rho = 0.3$; only the shape changes. The uniform branch uses $U(-\sqrt{3}, \sqrt{3})$; the t_5 branch samples with covariance $\frac{5}{3} \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$ and divides by $\sqrt{5/3}$; the χ^2_2 branches center by 2 and scale by $\sqrt{4}$. `chisq_same` leaves both margins right-skewed, whereas `chisq_diff` samples at $-\rho$ and negates W , flipping its skew while restoring $\text{Cor}(X, W) = +\rho$. Because $\text{Var}(XW)$ is a fourth-moment quantity, it is not constant across these conditions (≈ 1.09 under normality, larger under t_5 and χ^2).

Calibrating the misspecification magnitudes

The “small” and “medium” labels are standardized coefficients, $t \in \{0.25, 0.50\}$, not raw ones. For an omitted predictor p added to a baseline outcome y_0 , the magnitude is the fully standardized regression coefficient of p ,

$$t = \frac{\beta \text{sd}(p)}{\text{sd}(y_0 + \beta p)}.$$

Squaring and substituting $\text{Var}(y_0 + \beta p) = \text{Var}(y_0) + \beta^2 \text{Var}(p) + 2\beta \text{Cov}(p, y_0)$ yields a quadratic $A\beta^2 + B\beta + C = 0$ with

$$A = \text{Var}(p) (1 - t^2), \quad B = -2t^2 \text{Cov}(p, y_0), \quad C = -t^2 \text{Var}(y_0).$$

Since $A > 0$ and $C < 0$ the roots have opposite sign; the positive omitted path is the positive root,

$$\beta = \frac{t^2 \text{Cov}(p, y_0) + \sqrt{t^4 \text{Cov}(p, y_0)^2 + t^2 (1 - t^2) \text{Var}(p) \text{Var}(y_0)}}{\text{Var}(p) (1 - t^2)}.$$

The three moments are estimated from one large ($n = 10^6$) sample per distribution and a_3 level. The predictor p and baseline y_0 differ by misspecification,

$$c2 : \quad p = W, \quad y_0 = bM + c'X + \varepsilon_Y,$$

$$c3 : \quad p = XW, \quad y_0 = bM + c'X + \varepsilon_Y,$$

$$c1 : \quad p = X, \quad y_0 = m_3 = \lambda_3 M + \delta_{m_3}.$$

c2 and c3 are reliability-independent; c1 depends on ω because its baseline m_3 carries measurement noise. rc never enters the calibration: it is a residual correlation and equals its target in every cell.

Holding Y reliability fixed under c2 and c3

Adding an omitted structural path inflates $\text{Var}(Y)$ by $\kappa = \text{Var}(y_0 + \beta p) / \text{Var}(y_0)$, which would raise Y 's indicator reliability as a side effect. Scaling the Y -indicator noise by the same κ makes signal and noise move together, so κ cancels and the reliability returns to its no-misspecification value,

$$\omega_j = \frac{\kappa \lambda_j^2 V}{\kappa \lambda_j^2 V + \kappa \theta_j} = \frac{\lambda_j^2 V}{\lambda_j^2 V + \theta_j}.$$

The latent effect size is untouched; only the Y indicators are made correspondingly noisier. The measurement-level misspecifications c1 and rc act on the indicators directly and are not compensated, since that contamination is itself the misspecification.